

Taxonomy of Multimodal Time Series Models

Multimodal Time Series Paradigm (Data, Fusion, Role, Task, Domain)

1. Modality Pairing

- TS + Text (Reports, News, Prompts)
- TS + Vision (Plots, Spectrograms, MAE)
- TS + Proprioception / Telemetry (ACT, HiPolicy)
- TS + Quantum Circuits (Quantum-Mamba)
- TS + Spikes & Wireless RF (NeuromorphicSplit)
- TS + Physics & Graphs (PhysDGM, H-STQGCN)

2. Fusion Architecture

- Patch Reprogramming (Time-LLM, OFA)
- Action Chunking & Diffusion (ACT, HiPolicy)
- Quantum-Classical SSM (Quantum-Mamba)
- Semantic Rate-Distortion Coding (SemanticTS)
- Continuous Neural ODE / CDE (LLMODE, SOTER)
- Federated Sparse Adaptation (PerFed-TSFM)

3. Non-TS Role

- Auxiliary Context / Condition
- Embodied Visuomotor Goal Anchor
- Quantum Entangled Topology Prior
- Goal-Oriented Semantic Rate Control
- Formal Invariant Certificate (SELA)
- Metric Alignment Target (TRACE)

4. Downstream Tasks

- Multimodal Forecasting (Point / Conformal)
- Embodied Robotic Manipulation & Control
 - Quantum Spatio-Temporal Prediction
- Distributed Edge-Cloud Split Inference
- Spatio-Temporal Reasoning & Causal QA
- Safety-Critical CPS Logic Verification

Application Domains: Robotics & Manipulation • Healthcare & EHR • Meteorology & Climate • Finance & Markets • Energy & Smart Grids • Industrial IoT & 6G